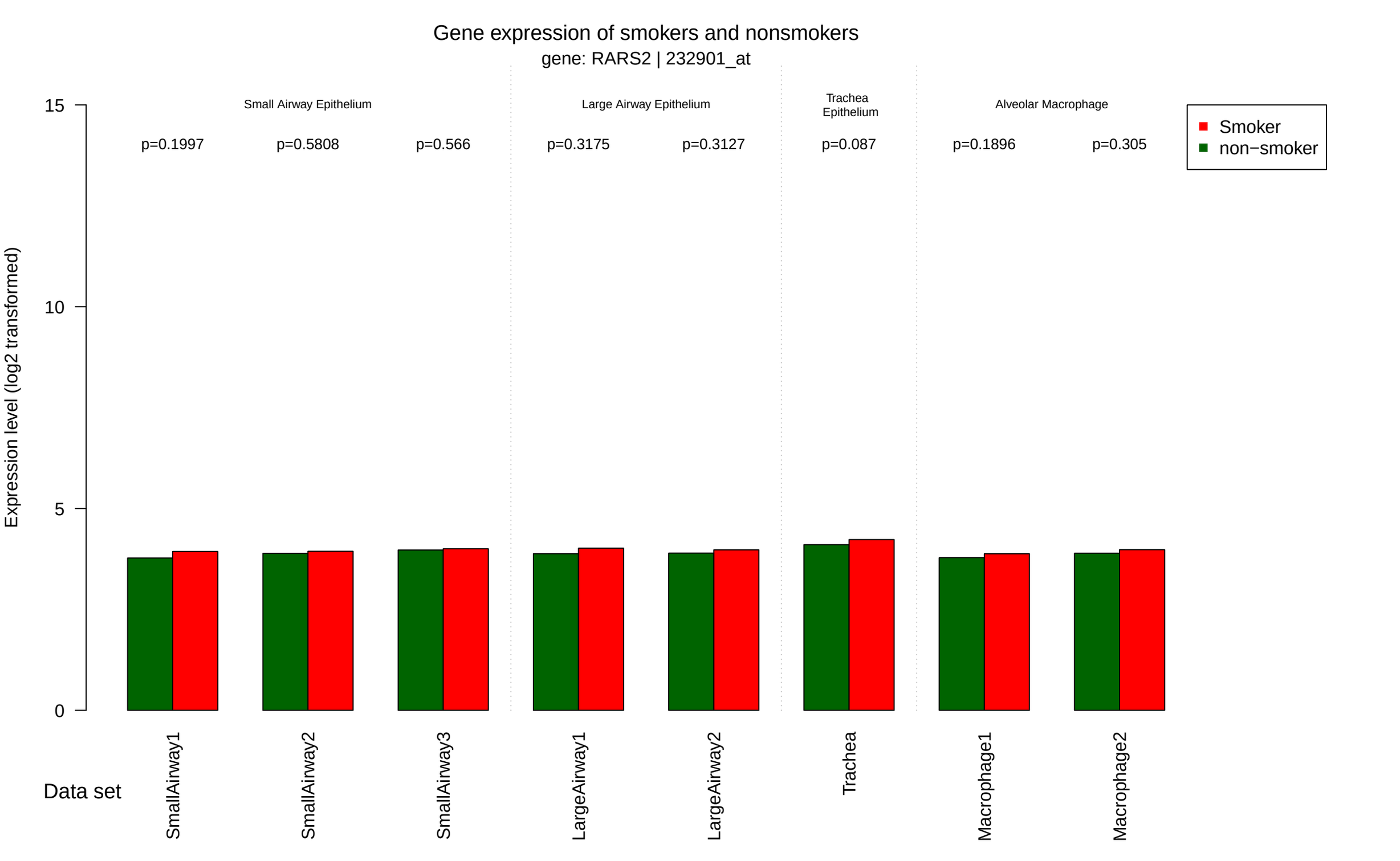
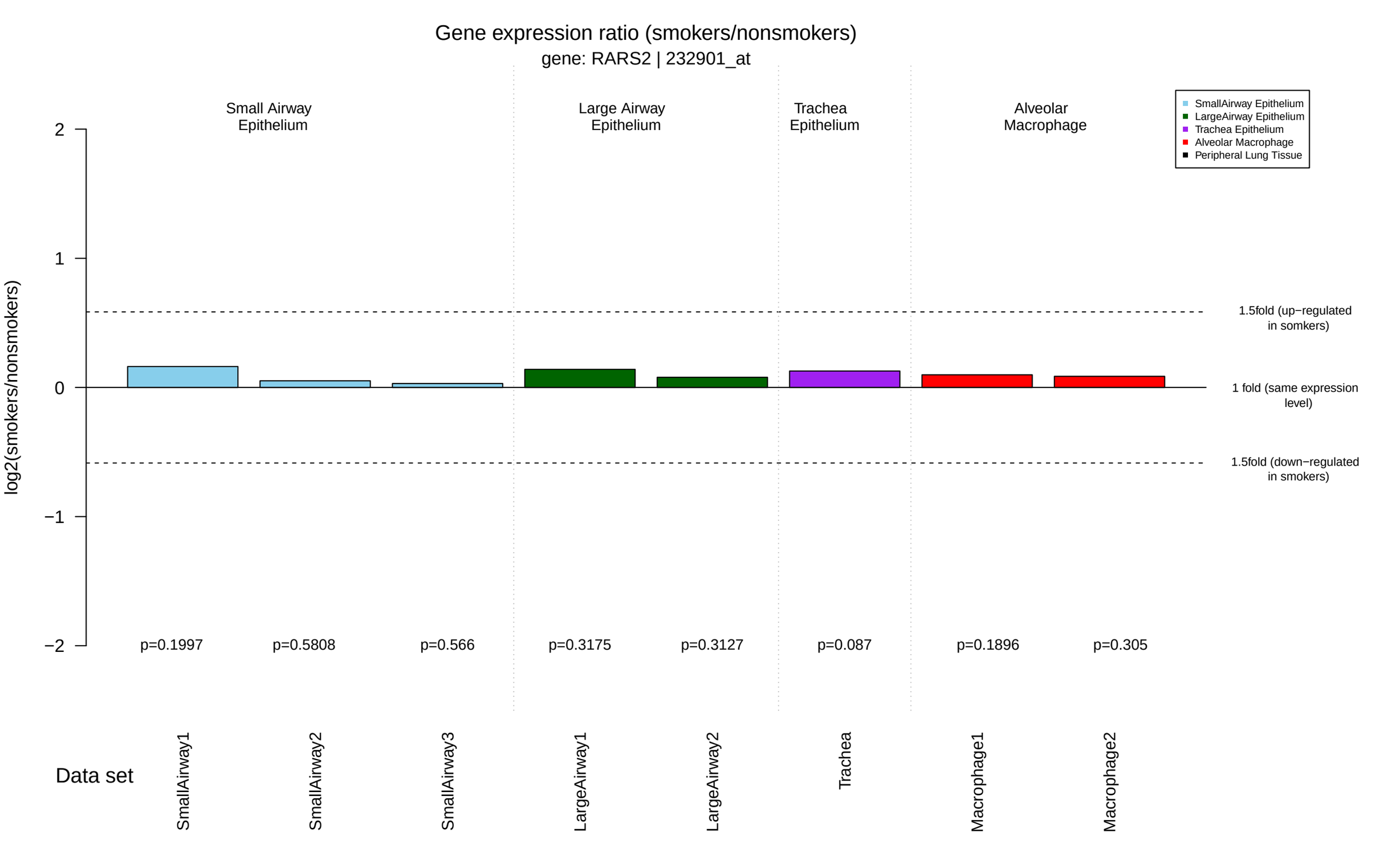
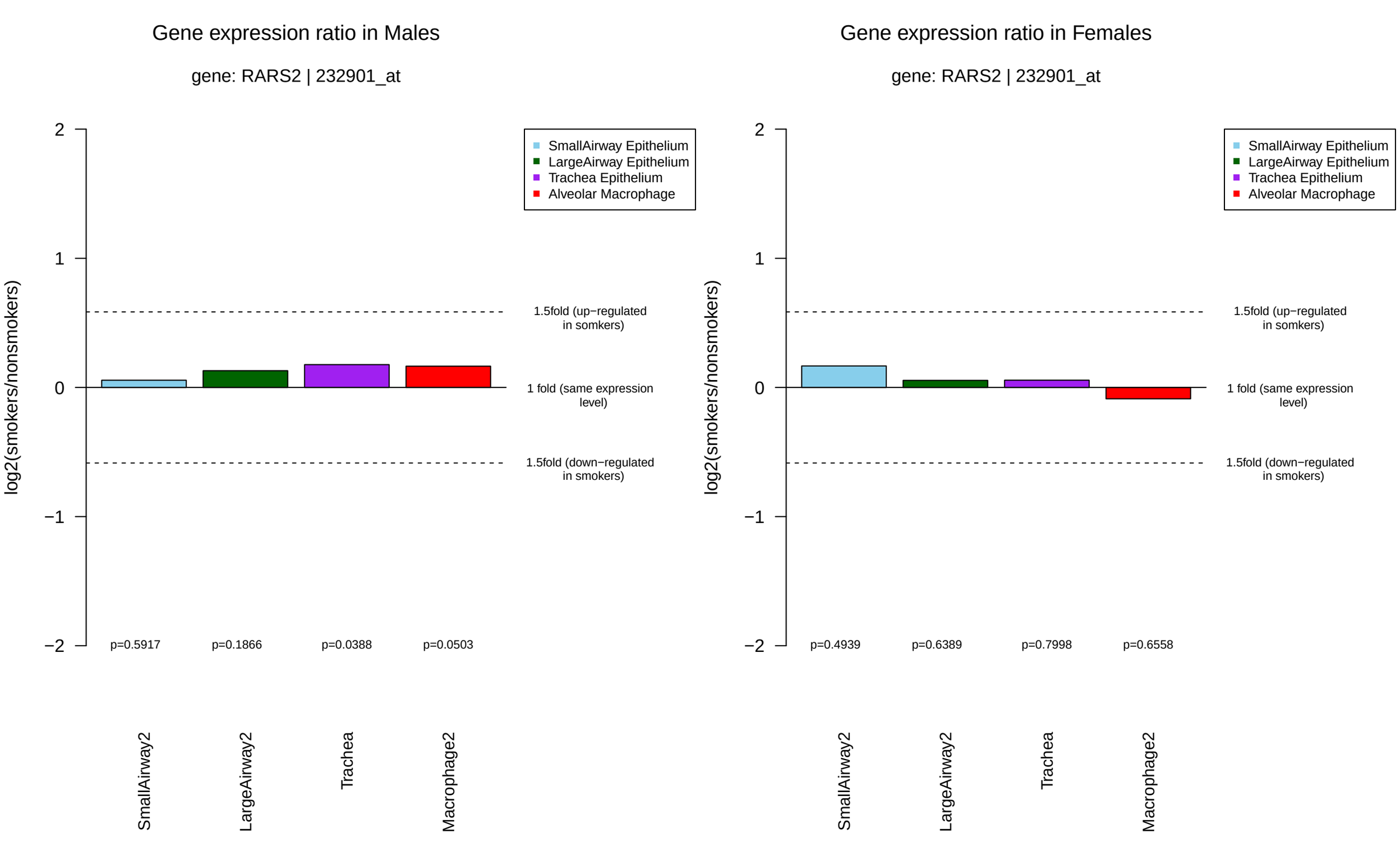
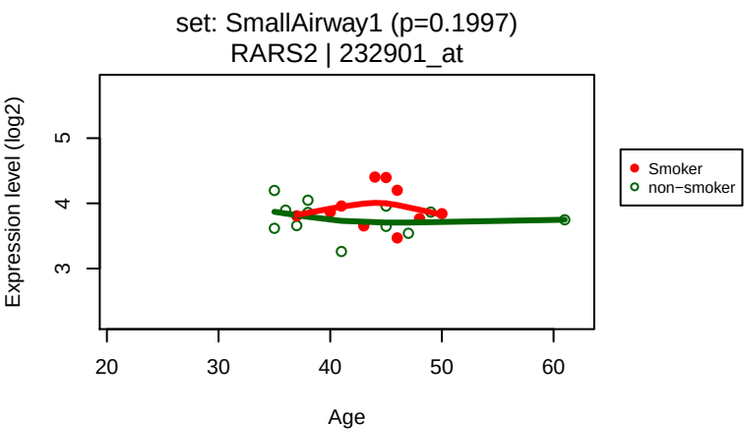
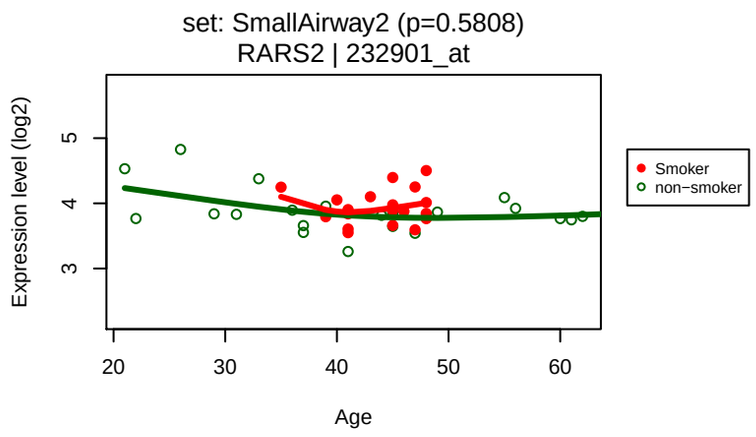
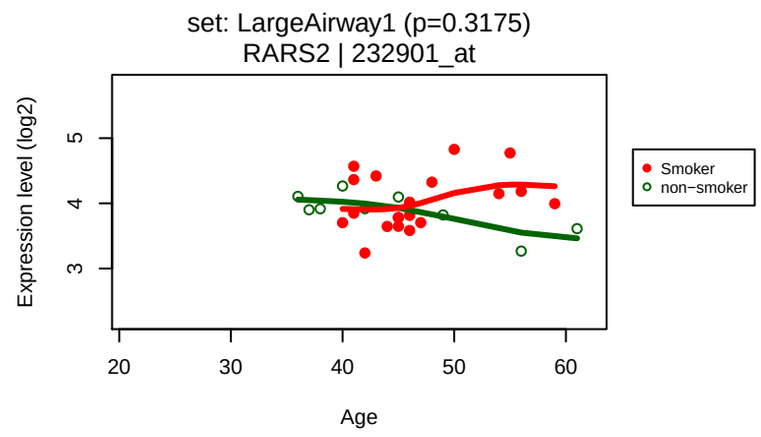

Gene expression of smokers and nonsmokers
gene: RARS2 | 232901_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1997
p=0.5808
p=0.566
p=0.3175
p=0.3127
p=0.087
p=0.1896
p=0.305
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

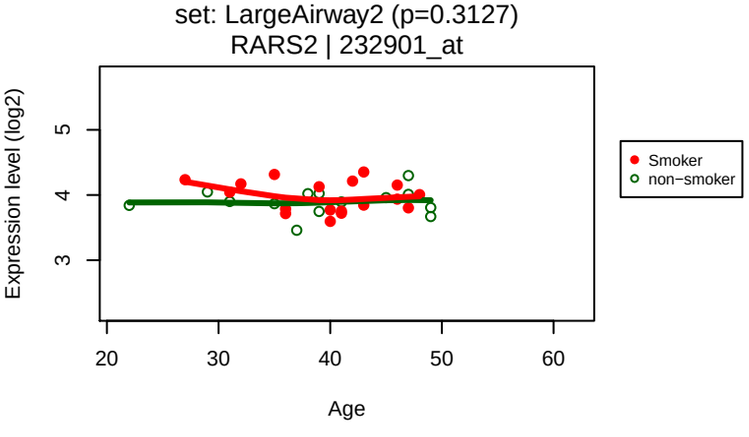

Gene expression ratio (smokers/nonsmokers)
gene: RARS2 | 232901_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1997
p=0.5808
p=0.566
p=0.3175
p=0.3127
p=0.087
p=0.1896
p=0.305
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

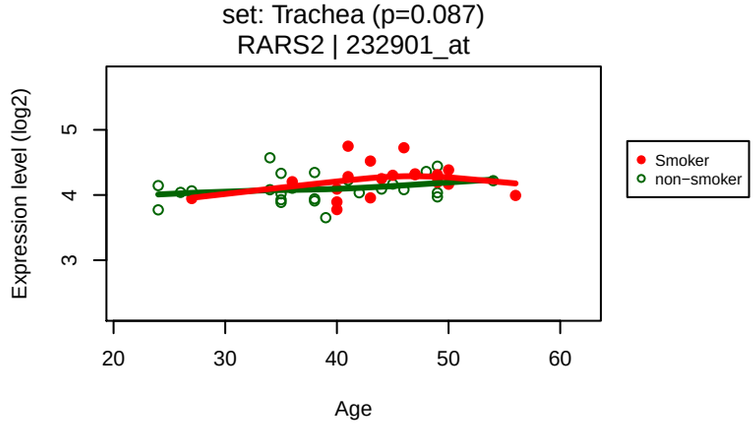

Gene expression ratio in Males
gene: RARS2 | 232901_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5917
p=0.1866
p=0.0388
p=0.0503
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RARS2 | 232901_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4939
p=0.6389
p=0.7998
p=0.6558
SmallAirway2
LargeAirway2
Trachea
Macrophage2

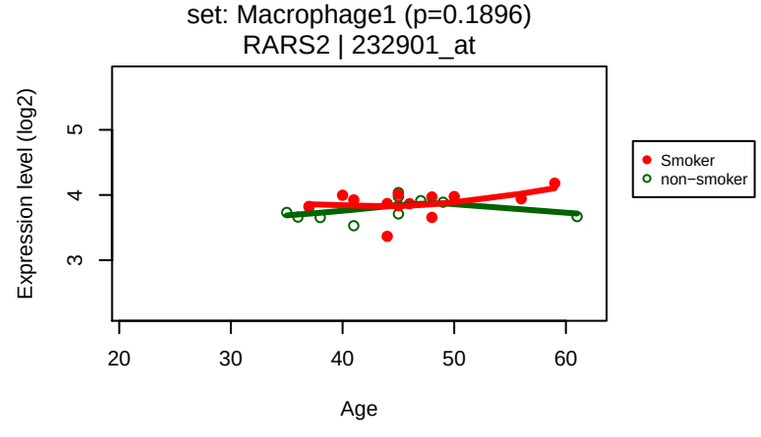

set: SmallAirway1 (p=0.1997)
RARS2 | 232901_at
Expression level (log2)
Age
Smoker
non−smoker

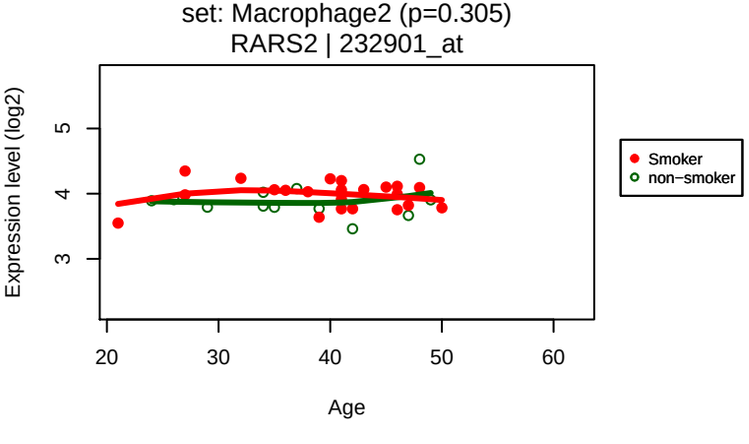

set: SmallAirway2 (p=0.5808)
RARS2 | 232901_at
Expression level (log2)
Age
Smoker
non−smoker

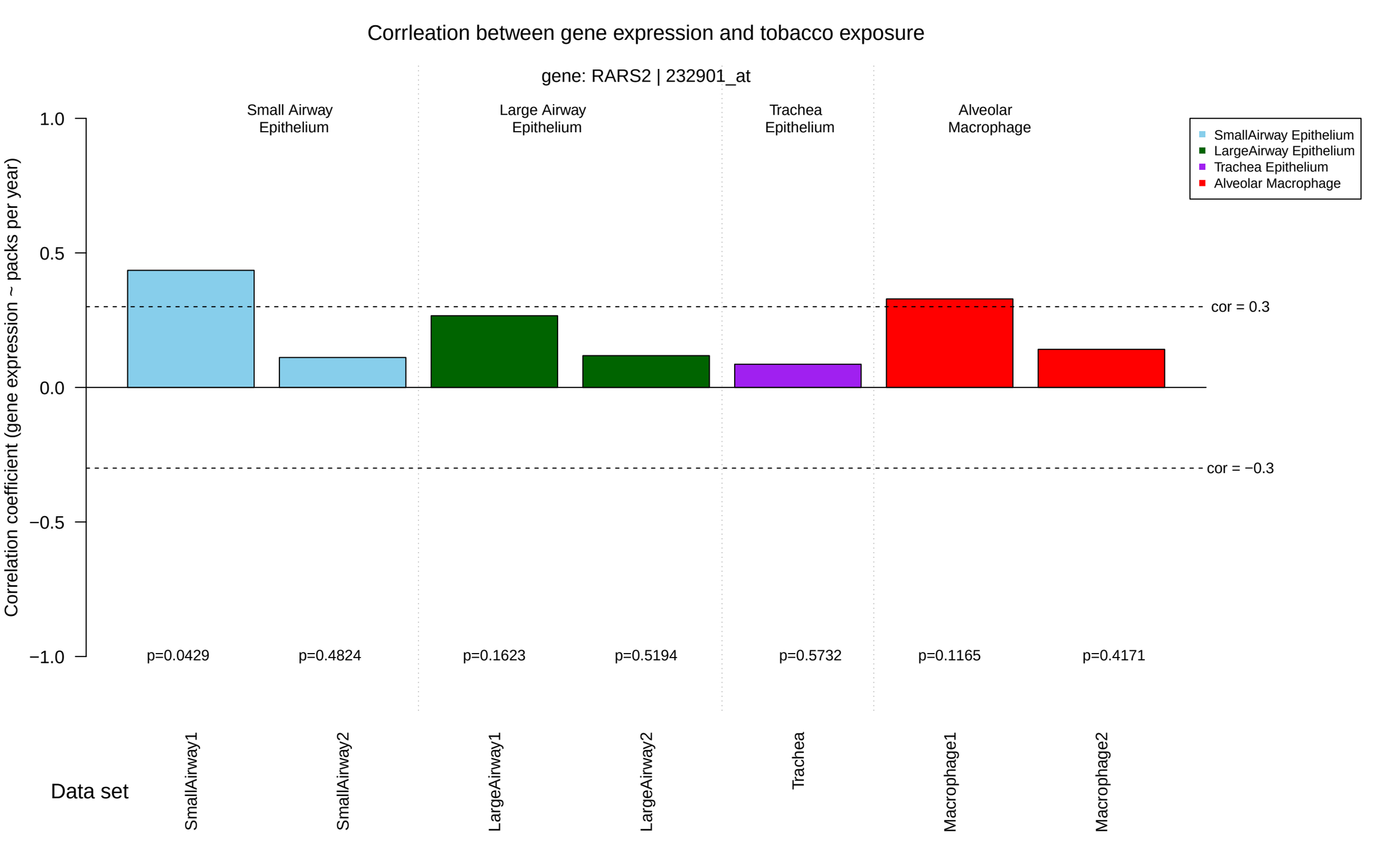

set: LargeAirway1 (p=0.3175)
RARS2 | 232901_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3127)
RARS2 | 232901_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.087)
RARS2 | 232901_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1896)
RARS2 | 232901_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.305)
RARS2 | 232901_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RARS2 | 232901_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0429
p=0.4824
p=0.1623
p=0.5194
p=0.5732
p=0.1165
p=0.4171
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2